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Car License Plate Identification using Jetson Nano

Literature Review

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Introduction

Literature Review Article: 1

Topic 1: A New CNN-Based Method for Multi-Directional Car License Plate Detection I

Description:
This paper conducts a comparative analysis of different object recognition methods, including RCNN, Faster RCNN, and YOLO. The study reveals that YOLO outperforms other object detection techniques in terms of both speed and accuracy. RCNN and Faster RCNN, characterized by numerous training parameters and complex neural network structures, face challenges in real-time identification. In contrast, YOLO divides the image into grids of size S x S, predicting bounding boxes and classes for each grid, eliminating the need for candidate regions. This simplification leads to a significant enhancement in detection speed.(**Xie:2018**)
Keywords:
Machine learning, YOLO, CNN, Real time systems , Deep learning
Conclusion

Topic 1: A New CNN-Based Method for Multi-Directional Car License Plate Detection II

This source was chosen for its detailed comparison of object detection methods, highlighting YOLO's speed and accuracy, but it lacks quantitative results and overlooks potential practical limitations.

Literature Review Article: 2

Topic 2: YOLO Algorithm Implementation for Real Time Object Detection and Tracking I

Description:
This paper investigates a deep learning method using YOLO for robust vehicle license plate detection and character recognition in diverse environments. It excels in detecting plates and accurately recognizing characters, but faces challenges like plate tilting, light reflection, diverse lighting, rain, and night conditions, impacting recognition accuracy.(**Garvi:2022**)
Keywords:
Deep learning, License plate, Recognition, Intelligent transportation system
Conclusion
This source was selected for its exploration of deep learning-based license plate recognition with the YOLO model, showcasing robust detection and character recognition. It effectively covers the method's strengths but

Topic 2: YOLO Algorithm Implementation for Real Time Object Detection and Tracking II

acknowledges its limitations in handling external factors, such as tilting, reflections, and adverse weather conditions.

Literature Review Article: 3

Topic 3: Automatic recognition of car license plates based on deep learning for convergence traffic control system I

Description:
This paper offers a brief survey of five key object detection methodologies: YOLO, ConerNet, R-CNN, SPPNet, and Auto-FPN. Object detection involves identifying objects in images, categorizing them, and determining their locations. This task is a significant challenge in computer vision due to variations in object attributes like shape, size, pose, color, and environmental factors, making it a formidable problem in the field.(**Ahn:2023**)
Keywords:
Object Detection, Convolutional Neural Network, Object detection algorithms, Accuracy, Performance
Conclusion
This source was chosen for its in-depth examination of five notable object detection methods (YOLO, ConerNet, R-CNN, SPPNet, Auto-FPN) and

Topic 3: Automatic recognition of car license plates based on deep learning for convergence traffic control system II

insights into object detection challenges, though it might not feature the latest developments in the field.

Literature Review Article: 4

Topic 4: Object Detection Algorithms: A Comparison I

Description:
This article explores computer vision object detection, categorizing it into one-stage and two-stage detectors. One-stage detectors like YOLO provide speed, while two-stage detectors (e.g., ConerNet and R-CNN) offer accuracy through region proposals. It assesses their pros and cons, emphasizing speed and complexity, and summarizes five key works (YOLO, ConerNet, R-CNN, SPPNet, Auto-FPN) to provide insights for those interested in computer vision and object detection methods. A valuable read for enthusiasts in this field.(**Shi:2022**)
Keywords:
Object detection ,YOLO, CNN,deep learning
Conclusion
This paper summarizes object detection methods with strengths, adaptability, and potential use in 3D point cloud applications.

Literature Review Article: 5

Topic 5: Comparison of Faster-RCNN, YOLO, and SSD for Real-Time Vehicle Type Recognition I

Description:
This article delves into deep learning models for vehicle type recognition, assessing Faster-RCNN, YOLO, and SSD for real-time processing and accuracy. Trained on automotive data, YOLOv4 outshines the others with 93 percent accuracy, thanks to FPN utilization and efficient object detection. Even with limited GPU resources, it stands out as the ideal choice.(**Kim:2020**)

Keywords:
Deep Learning , Object Detection , Real-time processing ,Yolo.

Conclusion
The study evaluated object detection methods for vehicle recognition, with YOLOv4 excelling due to innovative FPN use, addressing small object detection. Data limitations suggest unbiased training and more resources for better results.

Literature Review Article: 6

Topic 6: Vehicle License Plate Recognition I

Description:
This paper focuses on vehicle license plate recognition (VLPR) and its critical role in applications such as law enforcement and commercial use. It explores the use of the YOLO (You Only Look Once) deep learning architecture, specifically YOLO version 5, to detect license plates in image datasets. The emphasis is on achieving precise detection within intelligent transportation systems and image recognition, offering insights into optimizing license plate recognition systems for improved efficiency and effectiveness in various domains, from law enforcement to commercial applications.(**Aman:2021**)

Keywords:
Vehicle License Plate Recognition , Deep Learning , Yolo, Image Recognition.

Conclusion
YOLOv5 improves object detection, particularly in vehicle recognition. Advantages include speed and user-friendliness in PyTorch.

Literature Review Article: 7

Topic 7:Automatic License Plate Recognition System Using SSD I

Description:
This paper presents a real-time Automatic License Plate Recognition (ALPR) system using deep learning-based plate localization and a comparative analysis of Tesseract OCR and Easy OCR, achieving over 95 percent accuracy on NVIDIA Jetson Nano. [(**Awalgaonkar:2021**)].
Keywords:
NVIDIA Jetson Nano, Image Acquisition, Deep Learning, Image Preprocessing, License Plate Localization.
Conclusion:
The proposed ALPR system showcases efficient license plate recognition and underscores the superiority of Easy OCR, making it a promising solution for applications like parking management and theft detection.

Literature Review Article: 8

Topic 8: Towards End-to-end Car License Plate Location and Recognition in Unconstrained Scenarios I

Description:
The paper proposes a novel framework for car license plate location and recognition in unconstrained scenarios, such as different weather conditions, illumination levels, occlusions, and viewpoints.The network is based on the YOLO model, which is a family of object detection models that can predict bounding boxes and class labels in one pass. The network consists of three parts: a backbone network, a feature pyramid network, and a detection head.(**Qin:2020**)
Keywords:
Convolutional neural networks, YOLO, Anchor-free method.
Conclusion
The paper also compares the proposed framework with other methods that use different YOLO models, such as YOLOv3 and YOLOv4. The paper shows that the proposed framework achieves comparable results

Topic 8: Towards End-to-end Car License Plate Location and Recognition in Unconstrained Scenarios II

with these methods while using a smaller network size and fewer parameters.

Literature Review Article: 9

Topic 9: Real-time Automatic License Plate Recognition System using YOLOv4 I

Description:
The paper presents a real-time Automatic License Plate Recognition (ALPR) system using YOLOv4, aimed at recognizing license plates under challenging conditions such as small or unclear images from high-speed moving vehicles. The system eliminates the need for a Region of Interest (ROI) setting step, reducing computational complexity and improving recognition performance.(**Sung:2020**)

Keywords:
License Plate Recognition, YOLOv4, Real-time System

Conclusion
The proposed ALPR system leverages YOLOv4 for small license plate detection and character recognition, achieving both speed and accuracy while processing field data. The authors suggest future work should include expanding the character recognition dataset to improve model performance.

Literature Review Article: 10

Topic 10: Detecting License Plate Number Using OCR Technique and Raspberry Pi 4 With Camera I

Description:
This paper introduces a car license plate detection and recognition system with four main stages: image processing, segmentation, noise removal, and deep learning. Utilizing a Raspberry Pi 4, Python language, and a camera, the system employs YOLOv4 algorithm and Cascade Classifier for accurate license plate detection. The YOLOv4 technique achieves a 0.999 accuracy, while the Cascade Classifier, with Russian and Indian number plate databases, achieves 0.982 and 0.906 accuracy, respectively. Extracted characters are processed using the easy OCR library, facilitating license plate recognition through Python, OpenCV, Numpy, and easy OCR libraries.(**Aljelaw:2022**)

Keywords:
License Plate Recognition, YOLOv4, Cascade Classifier, Image Processing, Segmentation, Deep Learning, Raspberry Pi 4, Python, OpenCV, Numpy, easy OCR.

Literature Review Article: 11

Topic 11: Understanding of a convolutional neural network I

Description:
This paper provides an in-depth exploration of Deep Learning, specifically focusing on Artificial Neural Networks (ANN) with multiple layers. Over the past decades, Deep Learning has emerged as a powerful tool capable of handling vast amounts of data. The shift towards deeper hidden layers, notably in Convolutional Neural Networks (CNNs), has demonstrated superior performance in various fields, particularly in pattern recognition. CNNs, named after the mathematical operation of convolution, comprise layers such as convolutional, non-linearity, pooling, and fully-connected layers. With parameters influencing convolutional and fully-connected layers, CNNs excel in machine learning problems, particularly those involving image data, computer vision, and natural language processing.(**Albawi:2017**)
Keywords:

Topic 11: Understanding of a convolutional neural network II

Deep Learning, Artificial Neural Networks, Convolutional Neural Network (CNN), Machine Learning, Image Data, Pattern Recognition, Computer Vision, Natural Language Processing (NLP).

Literature Review Article: 14

Topic 14: Deep Learning vs. Machine Learning – What's The Difference? I

Description:
The blog post from Levity.ai delves into the nuanced differences between machine learning and deep learning within the realm of artificial intelligence. It provides valuable insights into the unique characteristics and applications that define these two crucial branches, offering clarity to those navigating the landscape of AI technologies.(**Arne:2022**)
Keywords:
Machine Learning, Deep Learning, Artificial Intelligence, Distinctions, Applications

Literature Review Article: 15

Topic 15: Deep Learning: Methods and Applications I

Description:
The article "Deep Learning: Methods and Applicationsëxplores diverse methods and applications within the realm of deep learning. For a comprehensive understanding of the content, accessing the article directly is recommended.(**Deng:2014**)
Keywords:
Deep Learning, Methods, Applications

Literature Review Article: 16

Topic 16: EasyOCR - Reading Texts I

Description:
The documentation page ËasyOCR - Reading Textsünder Open eVision 2.5 provides detailed information on using EasyOCR for text recognition. It is recommended to refer to the documentation directly for a comprehensive understanding of the features and functionalities discussed.(**EasyOCR:2018**)
Keywords:
EasyOCR, Text Recognition, Open eVision 2.5, Documentation

Literature Review Article: 17

Topic 17: TensorFlow Lite Guide I

Description:
TensorFlow Lite is a lightweight and streamlined version of the TensorFlow machine learning framework, specifically designed for deploying models on mobile and embedded devices. It enables efficient execution of machine learning models on resource-constrained platforms, making it ideal for applications such as mobile apps, edge devices, and IoT.(**GoogleTensorFlow:2022**)
Keywords:
TensorFlow Lite, Machine Learning, Deployment, Mobile Devices, Embedded Systems, Edge Computing, IoT

Literature Review Article: 18

Topic 18: Image Recognition with Deep Learning I

Description:
Image recognition, a pivotal field in image processing and computer vision, encompasses diverse applications. This paper addresses the unique challenge of food image classification, an essential component for dietary assessment systems in an era where health consciousness prevails. The inherent non-linearity of food image datasets poses a significant challenge. To tackle this, we propose a method utilizing Convolutional Neural Networks (CNNs) for accurate food category classification.(**Islam:2018**)

Keywords:
Image Recognition, Food Image Classification, Convolutional Neural Network (CNN), Dietary Assessment, Computer Vision, Neural Networks, Object Detection, Health Consciousness.

Literature Review Article: 19

Topic 19: Unsupervised Machine Learning I

Description:
The content on the provided link delves into the realm of Unsupervised Machine Learning. It covers concepts where algorithms autonomously identify patterns and structures within unlabeled data. For a detailed exploration, it is recommended to visit the link directly.(**JavaTPoint:2021**)
Keywords:
Unsupervised Machine Learning, Machine Learning, Clustering, Dimensionality Reduction, Anomaly Detection

Literature Review Article: 20

Topic 20: Deformation models for image recognition I

Description:
This study explores the application of diverse nonlinear image deformation models in the context of image recognition, focusing on their efficacy in addressing local variations commonly encountered in object variability. Among the models considered, a particular approach stands out for its simplicity in implementation, low computational complexity, and remarkable performance across various real-world image recognition tasks.(**Keysers:2007**)
Keywords:
Image Recognition, Nonlinear Deformation Models, Handwritten Digit Recognition, Medical Image Classification, Generalization Capacity, MNIST Benchmark, ImageCLEF Evaluation.

Literature Review Article: 21

Topic 21: An introduction to deep learning I

Description:
Deep learning, known for automatically extracting hierarchical representations from data, is examined in this study, focusing on a particular implementation—stacked denoising autoencoders. The research contributes by showcasing that utilizing this representation, in conjunction with a Support Vector Machine (SVM), enhances classification accuracy on the MNIST dataset. This improvement is observed in comparison to the conventional deep learning approach, where a logistic regression layer is typically added to the stack of denoising autoencoders.(**Lauzon:2012**)

Keywords:
Deep Learning, Stacked Denoising Autoencoders, Support Vector Machine, MNIST Dataset, Classification Error, Representation Learning, Logistic Regression.

Literature Review Article: 22

Topic 22: Decision trees in Machine Learning I

Description:
The article on Towards Data Science delves into the realm of decision trees in machine learning. It is anticipated to provide comprehensive information on the principles, applications, and implementation of decision trees, particularly in the context of classification and regression within supervised learning. For an in-depth exploration, it is recommended to visit the link directly.(**Prashant:2017**)
Keywords:
Decision Trees, Machine Learning, Classification, Regression, Supervised Learning

Literature Review Article: 23

Topic 23: What is Random Forests? I

Description:
The blog post on CareerFoundry delves into the topic of Random Forest in the realm of data analytics and machine learning. It is expected to offer clarity on the definition, applications, and significance of Random Forest, particularly within the context of ensemble learning. For a detailed understanding, it is recommended to explore the content directly through the provided link.(**RachaelMeltzer.2021**)
Keywords:
Random Forest, Data Analytics, Machine Learning, Ensemble Learning

Literature Review Article: 24

Topic 24:Python machine learning: Machine learning and deep learning with Python, scikit-learn, and TensorFlow / Sebastian Raschka & Vahid Mirjalili I

Description:
The book Python Machine Learningäuthored by Sebastian Raschka and Vahid Mirjalili is a comprehensive guide that explores machine learning and deep learning using Python, scikit-learn, and TensorFlow. The content likely covers fundamental concepts, practical applications, and hands-on examples using popular Python libraries. This resource can serve as a valuable reference for both beginners and experienced practitioners in the field of machine learning.(**Raschka:2017**)
Keywords:
Python, Machine Learning, Deep Learning, scikit-learn, TensorFlow, Data Science, Programming, Artificial Intelligence.

Literature Review Article: 25

Topic 25:You Only Look Once: Unified, Real-Time Object Detection I

Description:
This paper introduces YOLO (You Only Look Once), a novel approach to object detection that reframes the problem as a regression task for predicting spatially separated bounding boxes and associated class probabilities. YOLO utilizes a single neural network to directly predict bounding boxes and class probabilities from entire images in a single evaluation, allowing for end-to-end optimization on detection performance. The unified architecture of YOLO is exceptionally fast, with the base model processing images in real-time at 45 frames per second. Even the smaller version, Fast YOLO, achieves an impressive 155 frames per second while maintaining double the mean average precision (mAP) of other real-time detectors.(**Redmon:2016**)

Keywords:
YOLO, Object Detection, Neural Networks, Regression, Real-Time Detection, mAP, Generalization, DPM, R-CNN.

Literature Review Article: 26

Topic 26:Automatic Car Number Plate Recognition System for Authorization I

Description:
In the current context, ensuring security and authentication is paramount for creating a safe environment. Automatic License Plate Recognition (ALPR) emerges as a crucial application for authorization, alleviating the need for constant human monitoring. This proposed method addresses the challenge through a three-fold process: firstly, the extraction of the number plate region; secondly, the segmentation of characters; and finally, authorization through recognition and classification.(**Sandha:2017**)
Keywords:
Automatic License Plate Recognition, Security, Authentication, Morphological Approaches, Neural Network, Recognition, Classification, Authorization.

Literature Review Article: 27

Topic 27:Automatic Car Number Plate Recognition System for Authorization I

Description:
In the current context, ensuring security and authentication is paramount for creating a safe environment. Automatic License Plate Recognition (ALPR) emerges as a crucial application for authorization, alleviating the need for constant human monitoring. This proposed method addresses the challenge through a three-fold process: firstly, the extraction of the number plate region; secondly, the segmentation of characters; and finally, authorization through recognition and classification.(**Sandha:2017**)
Keywords:
Automatic License Plate Recognition, Security, Authentication, Morphological Approaches, Neural Network, Recognition, Classification, Authorization.

Literature Review Article: 28

Topic 28:Image recognition based on deep learning I

Description:
Image recognition based on deep learning has emerged as a powerful and versatile approach in the field of computer vision. Leveraging the capabilities of deep neural networks, this methodology enables automated and accurate identification of objects, patterns, or features within images. The deep learning models, particularly convolutional neural networks (CNNs), are designed to learn hierarchical representations from data, allowing for robust and efficient image recognition.(**Wu:2015**)

Keywords:
Image Recognition, Deep Learning, Convolutional Neural Networks, Computer Vision, Object Detection, Classification, Scene Understanding.

Literature Review Article: 29

Topic 29: Real-Time Automatic License Plate Detection and Blurring Using the European Union Dataset I

Description:
This paper presents the THI License Plate Dataset (TLPD), a large EU dataset with over 17,000 vehicle images and 18,000 labeled license plates. It covers varied angles, distances, lighting, and weather conditions to support robust license plate detection. A fast annotation tool enables efficient labeling. Two detection networks trained on TLPD achieved 93% precision and recall under normal conditions and 87% in heavy snow. Motivated by European data protection laws, the work supports real-time license plate detection and blurring for privacy protection.(**Chan:2023**)
Keywords:
License Plate Detection, European Dataset, Annotation Tool, Machine Learning, Deep Learning, Computer Vision, Real-Time Processing, Privacy Protection, Intelligent Transportation Systems.

References

References I

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Thanks a lot

for your attention